An Agentic Framework for Social Event Forecasting: Approaches using Causality Contextualized Chain of Thought

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***Abstract*—Recent advances in large language models (LLMs) have enabled impressive progress across diverse tasks, yet interpretability remains a core requirement for deployment in high-stakes domains such as crisis prevention and policy-making. Prior work on event prediction has largely prioritized accuracy, but the reasoning behind model outputs often remains opaque and difficult to audit. In this paper, we propose *C3OT*, Causality Contextualized Chain-of-Thought, which integrates causal reasoning into an agentic LLM framework using the ReAct paradigm. We design and evaluate multiple prompting strategies—including Causal Chain Learning, Chain-of-Thought, and more nuanced hybrid approaches. Experiments assess both predictive accuracy and interpretability, the latter measured through structured rubrics that capture transparency, causal coherence, and auditability. Results demonstrate that our causal reasoning approach attains competitive predictive performance while producing more transparent and auditable reasoning traces. These findings underscore the value of causal reasoning for enhancing both trustworthiness and robustness in socio-political forecasting.**

***Index Terms*—Large Language Models, Agentic Workflows, Causal Reasoning, Chain-of-Thought, Event Prediction**

I. Introduction

Interpretability constitutes a foundational requirement for the responsible deployment of application-specific use cases driven by large language models (LLMs) in domains where decisions carry substantial real-world implications [1], [20]. In high-stakes contexts such as policy-making, crisis response, or international conflict monitoring, predictive accuracy alone is inadequate. Forecasts must be accompanied by transparent reasoning that allows decision-makers to evaluate the basis for a model's output before incorporating it into critical deliberations. Absent such interpretability, AI-assisted forecasts face the risk of being disregarded, misapplied, or distorted, thereby undermining their intended utility [12], [21]. Large language models (LLMs) excel at integrating structured and unstructured data, generating coherent reasoning, and retrieving external knowledge. Yet their explanations often remain weakly tied to the underlying decision process, leaving them as "black boxes" that undermine trust in critical workflows.

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Causal reasoning provides a principled remedy by requiring models to articulate explicit cause–effect relationships before producing outputs, thereby creating transparent, inspectable decision paths [2], [3], [18]. In high-stakes settings, such transparency is often more valuable than marginal gains in predictive accuracy. Socio-political event forecasting offers a natural testbed: it demands reasoning over complex causal chains, the integration of contextual and structured data, and carries significant risks if misinterpreted. This work adds a wrinkle to the event forecasting landscape by evaluating how incorporating causal reasoning enhances the interpretability and practical utility of LLM outputs in such contexts.

To address this, this study incorporates causal reasoning into an agentic framework based on the ReAct (Reasoning–Acting–Observing) paradigm [23]. This enables the model to iteratively gather evidence from structured temporal knowledge graphs and unstructured text sources, reason through causal chains, and produce predictions with explicit intermediate steps. The framework is tested on multiple prompt designs, including Causal Chain Learning (CL), Chain-of-Thought (CoT), Commonsense Reasoning (CSR), and C3OT (proposed hybrid approach), to analyze how prompt structure influences the interpretability of outputs [3], [6], [19], [22].

The contributions of this paper are threefold: (1) a framework for interpretable LLM outputs by incorporating causal reasoning into an agentic workflow, (2) evaluation of prompt design choices on explanation quality and traceability, and (3) a case study in socio-political forecasting to demonstrate the value of interpretability in sensitive prediction tasks.

II. Related Work

Recent scholarship highlights both the opportunities and limitations of interpretability in LLMs. They can generate natural-language explanations beyond post-hoc tools like feature attribution, but challenges such as hallucinations, opacity, and high computational costs remain, with priorities including dataset-level analysis and interactive workflows [21]. Shojaee et al. show that Large Reasoning Models (LRMs) succeed on simple to moderately complex tasks but collapse with higher complexity, suggesting reliance on pattern-matching

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over true reasoning [4]. Similarly, Huang et al. review prompting and training methods such as Chain-of-Thought and rationale engineering, concluding that while these improve performance, LLMs still struggle with multi-step reasoning and need reasoning-oriented architectures [20].

CausalEval introduces a benchmark for evaluating causal reasoning, distinguishing between LLMs as engines and helpers while surveying methods like Causal Chain-of-Thought (CausalCoT), Causal Contextualized Learning (CausalCL), and Multimodal Causal Reasoning (MCR). Although prompting and fine-tuning improve results, LLMs still lack deep causal understanding and require new architectures for robust inference [3]. Chain-of-Thought (CoT) Prompting improves reasoning in arithmetic, commonsense, and symbolic tasks by guiding models through structured intermediate steps [19]. Building on this, CLadder introduces Causal Chain-of-Thought (CausalCoT) for graph-based reasoning, showing gains but persistent inference errors [18]. Causal Contextualized Learning (CausalCL) reinserts reasoning steps into evolving context to maintain continuity, improving multi-step reasoning when context remains coherent [22]. Finally, Multimodal Causal Reasoning (MCR) identifies causal features while suppressing confounders across modalities, achieving state-of-the-art VideoQA performance [6].

Agentic frameworks structure reasoning more effectively. ReAct interleaves reasoning with tool calls and evidence gathering, improving transparency and factual grounding over pure CoT. Extensions like ART (automatic multi-step reasoning + tools) and CoH (Chain-of-History temporal graph reasoning) further generalize this approach, though faithfulness remains a challenge. These methods highlight a broader trend toward agentic designs that integrate planning, memory, and reflection, creating workflows closer to human-style reasoning. However, studies like C2P show that without explicit causal scaffolds, LLMs still risk confusing correlation with causation, motivating integration of causal prompts.

Socio-political event prediction provides an ideal testbed since understanding reasoning is essential for trust. While dynamic-graph and temporal-KG models (e.g., Know-Evolve; DyRep) achieve accuracy, they lack transparency. Hybrid approaches like GenTKG, MM-Forecast, and MIRAI combine LLMs with retrieval and structured graphs, while LEAP frames prediction as a language understanding task. Despite these advances, models often remain opaque to non-experts, making it difficult to validate their predictions in practice. The present work positions ReAct-style workflows augmented with causal prompts as a way to maintain competitive performance while exposing reasoning pathways critical for high-stakes decision support.

III. Methodology

A. Data Description

This study uses the POLitical Event Classification, Attributes, and Types (POLECAT) dataset, a large-scale machine-coded corpus of socio-political interactions that extends ICEWS and applies the PLOVER ontology to extract structured events from multilingual news [26], [27]. Each entry encodes actor, recipient, event type, timestamp, and location, represented in quadruple form:

$$\langle Actor, EventType, Recipient, Timestamp \rangle \quad (1)$$

This format captures directed interactions over time, enabling construction of temporal knowledge graphs.

To improve consistency, the following preprocessing steps are applied: (i) consolidate events across sources and remove duplicates, (ii) normalize missing or inconsistent locations using automated geocoding and ISO codes, (iii) harmonize actor/recipient fields through hierarchical fallbacks, and (iv) discard records with invalid timestamps. Finally, event types are mapped to conflict–cooperation intensity scores, allowing stratified analysis from hostile (e.g., *assault, sanction*) to cooperative (e.g., *aid, consult*) interactions.

B. Agentic Workflow

The workflow is structured around the ReAct paradigm (Reasoning–Acting–Observing) [23], which enables iterative cycles of reasoning, tool use, and evidence integration. This ensures that predictions are grounded in both structured and unstructured data while maintaining interpretability.

Temporal Knowledge Graph (TKG) Construction. Events from the POLECAT dataset [27] are transformed into a temporal knowledge graph. In this representation, nodes correspond to socio-political actors, while edges denote directed event relations between actors. Each edge is annotated with an event type and timestamp, enabling chronological ordering of interactions. This structure provides the agent with access to both relational and temporal dependencies, which are critical for forecasting.

Tooling Environment. The agent accesses structured and unstructured context through tools for retrieving connections, event metadata, historical frequencies, actor–recipient histories, and external news.

Iterative Reasoning Process. Guided by causal reasoning prompts, the agent follows a ReAct-style [23] iterative loop. The model first analyzes the query to identify causal factors and plans retrieval actions. It then invokes one or more tools to gather structured or unstructured evidence, incorporating the results into its reasoning chain and updating the causal narrative. This process repeats until sufficient evidence is accumulated to support a prediction.

Validation and Consistency Checks. To ensure reliability, predictions are validated by cross-checking against temporal patterns to prevent logical inconsistencies, such as predicting outcomes before their antecedents. Additional filters based on event frequency and intensity are applied to reduce noise and enforce semantic consistency, ensuring that predictions align with historically plausible event patterns.

The figure 1 depicts the flow of the agent framework. The agent uses the ReAct [23] strategy to decide on the Thought, the appropriate set of tools to use and based on the results, the agent forms a causal graph for reasoning to generate the final predictions.

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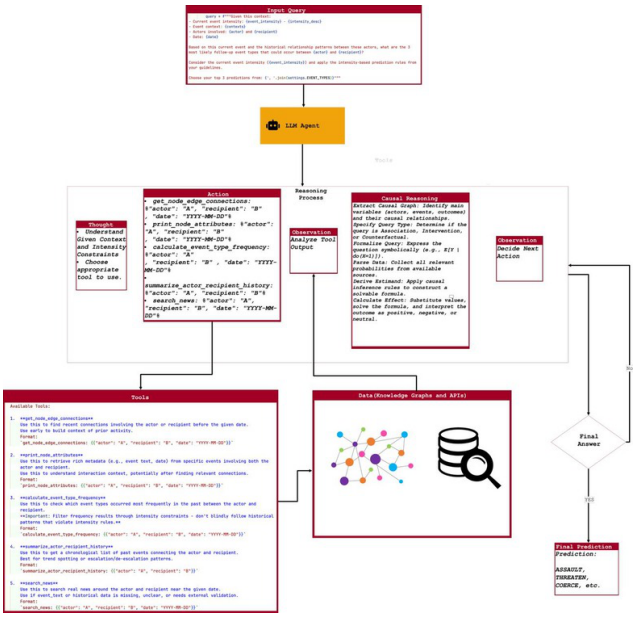


Fig. 1. Overview of the agentic workflow: The agent iteratively reasons, acts (via tool calls), and integrates evidence from both structured (temporal knowledge graph) and unstructured sources to generate interpretable predictions. A comprehensive overview of the system architecture can be found here: https://github.com/Adi-Sampath/AgentKG-Causal-Reasoning

C. Causal Reasoning and Prompt Design

Causal Chain-of-Thought (CausalCoT): CausalCoT extends traditional Chain-of-Thought prompting [5] by embedding causal inference principles into the reasoning process. Instead of producing a flat sequence of steps, the model is guided to extract causal graphs, classify query types (association, intervention, counterfactual), and formalize them into symbolic expressions before deriving an answer. Specifically, the process unfolds in six steps: extract a causal graph, identify the query type, formalize the query, parse available data, derive the estimand, and compute the answer. This structured approach encourages explanations that reflect cause-and-effect mechanisms rather than surface-level associations. To enforce this six-step process in our experiments, we designed custom prompts that explicitly instructed the model to follow each step in order, as illustrated in 1.

Causal Contextualized Learning (CausalCL): CausalCL builds on CoT by embedding reasoning within richer, context-aware prompts. After each reasoning step, newly inferred information is reinserted into the prompt, allowing the model to sustain context across multiple steps. This helps avoid the "forgetting" problem in long reasoning chains and promotes continuity in causal explanations.

Commonsense Reasoning: CSR can be viewed as a framework for strengthening commonsense reasoning by filtering out confounding information. Originally designed to integrate textual and visual modalities, Commonsense reasoning uses causal interventions to identify and suppress irrelevant or misleading features, forcing the model to attend only to signals

Algorithm 1 Causal Reasoning Prompt Structure (CausalCoT + ReAct)

1: **Extract the causal graph:** Identify main variables (actors, events, outcomes) and draw their causal links.
   Format: Graph = [list of edges, e.g., A $\rightarrow$ B, C $\rightarrow$ B]
2: **Identify the query type:** Choose from Association, Intervention, or Counterfactual.
   Format: Query Type = [your answer]
3: **Formalize the query:** Translate the question into a symbolic expression.
   Example: $E[Y \mid do(X = 1)] - E[Y \mid do(X = 0)]$
   Format: Symbolic Query = [expression]
4: **Parse available data:** Extract all probabilities given in the prompt (marginal, conditional, or interventional).
   Format: Data = [list of probabilities with values]
5: **Derive the estimand:** Use causal inference rules (e.g., backdoor adjustment, do-calculus) to build a solvable formula.
   Format: Estimand = [formula with substitutions]
6: **Calculate the answer:** Plug in numbers from Step 4, solve the formula, and explain the result.
   Format: Calculation = [worked example]
   State whether the effect is positive, negative, or neutral.

that reflect true cause-and-effect. This reduces the tendency of LLMs to rely on superficial correlations and instead promotes more grounded reasoning. In our experiments, where no video data was available, we adapted MCR to operate entirely within text, treating different textual perspectives or contexts as "modalities." By alternating between these perspectives, the model was able to filter out confounders and focus on coherent causal chains, enabling more robust commonsense reasoning.

C3OT (Hybrid CoT + CL): This hybrid method combines Causal Chain-of-Thought reasoning with Contextualized Learning. In our setup, we first applied CausalCL to expand and reinforce the context, ensuring the model had all necessary information carried forward. After this expansion, CausalCoT was applied so that the step-by-step reasoning could leverage the enriched context, producing explanations that were both logically structured and context-sensitive.

These approaches enhance interpretability by making the reasoning process explicit along with encouraging the model to consider not just what might happen, but why it might happen, focusing attention on underlying mechanisms rather than surface correlations. By structuring outputs into causal steps, each stage of the inference can be inspected, validated, and, if necessary, challenged by human evaluators. This visible reasoning chain reduces the risk of accepting less reliable post hoc explanations and provides a more trustworthy foundation for using LLMs in sensitive, high-stakes domains.

IV. Results and Discussion

A. Experimental Setup

The experimental evaluation was conducted on three country-specific subsets from the POLECAT dataset [27],

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Afghanistan (164 instances), India (1,998), and Russia (454), to benchmark the agentic framework against state-of-the-art baselines such as LEAP [9], which uses the ICEWS dataset [26]. This setup enables a like-for-like comparison across frameworks, with POLECAT serving as an updated successor to ICEWS [27]. The primary evaluation focuses on event type extrapolation, while LEAP emphasizes object prediction; however, both operate within the same temporal reasoning context and employ comparable datasets and metrics [9].

Performance was assessed using both quantitative and qualitative metrics. Quantitative metrics include Hit@1 (fraction of cases where the top-ranked prediction matches ground truth), Hit@3 (ground truth appears among top three predictions), and Mean Reciprocal Rank (MRR), which captures ranking quality by averaging the inverse rank of the first correct prediction. Qualitative metrics, adapted from the CausalEval framework [3] and normalized to a 0–1 scale, include Causal Graph Quality (accuracy of predicted causal links), Reasoning Coherence (logical consistency of multi-step reasoning), Context Utilization (effectiveness in leveraging temporal and historical context), and Error Typology (categorizing errors into logical, probabilistic, statistical, and contextual types).

B. Results

1) ***Overall Performance Comparison****:* The experimental set-up was tested using two state-of-the-art models GPT-4o-mini [16] and Gemini-2.5-pro [17] I summarizes quantitative results across three country subsets Afghanistan, India, and Russia, and prompt types (Chain Learning, Chain-of-Thought, Combined, Commonsense Reasoning).

Across all countries, the results demonstrate that prompt structure significantly influences predictive accuracy. Contextualized Learning (CL) prompts consistently yielded the highest Hit@1 and MRR scores, particularly for GPT-4o-mini, which achieved peak performance in the Russia subset (Hit@1: 70.26, MRR: 78.71). This suggests that explicitly maintaining context during reasoning enables models to better capture temporal and causal dependencies, leading to more reliable event predictions.

Performance varied notably by country. The Russia subset consistently produced the strongest results, while India posed the greatest challenge, likely due to higher event diversity and actor heterogeneity. Afghanistan results were intermediate, reflecting moderate complexity in event structures.

TABLE I
COUNTRY-WISE MODEL PERFORMANCE (CL, COT, C3OT, AND MCR)

Country	Prompt	Model	Hit@1 / Hit@3 / MRR
AFG	CL	GPT-4o-mini	68.29 / 87.80 / 76.73
	CL	Gemini-2.5-Pro	59.76 / 88.41 / 72.15
	CoT	GPT-4o-mini	68.90 / 83.54 / 75.41
	CoT	Gemini-2.5-Pro	46.34 / 79.88 / 61.08
	C3OT	GPT-4o-mini	62.80 / 83.54 / 71.54
	C3OT	Gemini-2.5-Pro	30.49 / 62.80 / 43.70
	CSR	GPT-4o-mini	59.76 / 73.17 / 65.85
	CSR	Gemini-2.5-Pro	24.39 / 62.80 / 41.26
IND	CL	GPT-4o-mini	51.70 / 74.87 / 61.82
	CL	Gemini-2.5-Pro	48.09 / 73.94 / 59.15
	CoT	GPT-4o-mini	49.15 / 71.97 / 58.95
	CoT	Gemini-2.5-Pro	33.00 / 68.80 / 47.80
	C3OT	GPT-4o-mini	50.20 / 66.87 / 57.32
	C3OT	Gemini-2.5-Pro	48.30 / 72.30 / 65.00
	CSR	GPT-4o-mini	49.71 / 67.79 / 57.85
	CSR	Gemini-2.5-Pro	19.30 / 45.61 / 30.95
RUS	CL	GPT-4o-mini	70.26 / 88.99 / 78.71
	CL	Gemini-2.5-Pro	54.50 / 89.50 / 69.17
	CoT	GPT-4o-mini	63.44 / 84.14 / 72.61
	CoT	Gemini-2.5-Pro	42.70 / 80.40 / 58.80
	C3OT	GPT-4o-mini	65.42 / 77.53 / 70.48
	C3OT	Gemini-2.5-Pro	42.47 / 80.25 / 78.26
	CSR	GPT-4o-mini	59.25 / 74.67 / 66.48
	CSR	Gemini-2.5-Pro	35.83 / 66.14 / 49.15

Hit@1, Hit@3, and MRR are reported as percentages.

2) ***Comparison with Baselines*** *:* To contextualize the efficacy of the proposed technique, we compared the proposed methodology against a basic ReAct agentic framework (using Mistral-7B and Qwen2.5-32B) and the LEAP framework [9].

TABLE II
COMPARISON WITH BASELINE FRAMEWORKS

Country	Framework	Hit@1	Hit@3	MRR
IND	GPT-4o-mini (proposed)	**51.70**	**74.87**	**61.82**
IND	Agentic Qwen-32B (Re-Act)	43.20	70.90	55.90
IND	LEAP (Fine-tuned) [9]	39.40	58.30	–
RUS	Gemini-2.5-Pro (proposed)	54.50	**89.50**	69.17
RUS	Agentic Qwen-32B	59.50	89.90	73.50
RUS	LEAP [9]	38.60	55.90	–

[a]Hit@1, Hit@3, and MRR are reported as percentages. "–" indicates not reported. The bold numbers highlight the best results of the proposed framework.

The proposed technique (with prompt-driven causal reasoning) achieves comparable or superior accuracy to Qwen-32B despite being lighter and consistently outperforms LEAP's fine-tuned approach in *Hit@k* [9]. These results highlight the advantage of integrating explicit causal reasoning into agentic frameworks, which not only delivers strong quantitative performance but also increases transparency and interpretability. Notably, performance varied by country: Russia exhibited the highest predictive scores and coherence, suggesting that more homogeneous event structures facilitate stable causal extraction, while India's lower performance likely reflects greater event diversity and actor heterogeneity, complicating causal inference. These findings underscore how dataset characteristics can significantly influence the effectiveness of causal reasoning frameworks.

3) ***Qualitative Results****:* The table III highlights the reasoning quality of the two models used in this study. This showcases how models can be evaluated on their reasoning capabilities and how models utilize the context for reasoning.

The experiments show that enforcing causal reasoning steps improves interpretability. GPT-4o-mini consistently achieved higher scores in graph creation and context utilization than Gemini, indicating clearer causal links and more effective use of temporal information. This reflects persistent gaps in decision support noted by CausalEval [3], where models often lacked transparent reasoning despite strong predictive

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TABLE III
COMPARISON OF MODELS ON QUALITATIVE REASONING METRICS ACROSS PROMPTS

Country	Model	Prompt	Graph Creation	Coherence	Context
RUS	GPT-4o-mini	CL	0.096	0.124	0.878
RUS	Gemini	CL	0.010	0.010	0.325
RUS	GPT-4o-mini	C3OT	0.547	0.557	0.341
RUS	Gemini	C3OT	0.000	0.002	1.000
RUS	GPT-4o-mini	CSR	0.000	0.987	0.554
RUS	Gemini	CSR	0.004	0.336	0.477

GPT-4o-mini demonstrates strong context utilization and reasoning coherence, especially with C3OT and CSR prompts. Gemini shows better context utilization, but struggles with qualitative consistency in complex prompts. Hybrid and Commonsense Reasoning strategies can enhance interpretability and reasoning depth beyond standard CL or CoT approaches.

accuracy. Overall, explicit causal prompting narrows the gap between raw outcome prediction and interpretable, trustworthy decision-making.

V. CONCLUSION AND FUTURE WORK

This study highlights the importance of prompt-driven causal reasoning within frameworks for improved interpretability for sensitive tasks such as socio-political event prediction [2], [3], [20]. By enforcing explicit causal steps and leveraging structured agentic workflows, the proposed approach consistently matches or outperforms baseline models, including fine-tuned frameworks [9], [12]. The results reveal that qualitative gains in graph creation, coherence, and context utilization are particularly pronounced when causal prompts are employed, especially in settings with more homogeneous event dynamics. However, performance remains sensitive to dataset complexity, with greater event diversity and actor heterogeneity posing ongoing challenges for stable causal extraction. These findings underscore the value of causal reasoning frameworks for high-stakes domains, where transparent decision support is as important as predictive power [3], [21]. Future work will explore scaling these methods to broader domains, multimodal and multilingual settings, as well as integrating long-term memory to learn reasoning and causal relation patterns over time, will be critical for advancing the reliability and adoption of interpretable LLM-based agents in real-world applications. Another promising direction is developing cross-agent causal reasoning, where multiple interacting models can coordinate to validate and refine each other's causal chains. In addition, interactive explanation interfaces could allow users to ask "why" or "what if" questions, providing deeper transparency into the reasoning process. Finally, incorporating human-in-the-loop validation would ensure that causal predictions remain grounded in domain expertise, creating a feedback loop that improves both model reliability and user trust.

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